Progressive exercises

E1 · Practice. In a frame of space, consider $A(1,-1,2)$ and $B(3,0,1)$. Give a parametric representation of the line (AB).

E2 · Practice. In an orthonormal frame, determine a Cartesian equation of the plane through $A(1,2,3)$ with normal vector $\vec{n} = (1,0,-2)$.

E3 · Practice. In a frame of space, the line d is given by $(x,y,z) = (t, 1+t, 2-t)$, $t \in \mathbb{R}$, and the plane $\mathcal{P}$ by $x + 2y + z = 8$. Determine their intersection.

E4 · Practice. In a frame of space, determine the intersection of the planes with equations $x + y + z = 1$ and $x - y = 0$. Give a parametric representation if the intersection is a line.

E5 · Practice. In a frame of space, study the relative position of the planes with equations $2x - y + z = 3$ and $4x - 2y + 2z = 5$. Justify.

Textbook or past-paper work

Baccalauréat, Metropolitan France, 17 June 2025: exercise 3, statements 1 to 4 (p. 5–6). At home after S70. Justify each answer; for a false statement, explain precisely what refutes it. The paper is available here: official past paper. Use the data and instructions of this past paper.